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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/698,915	04/05/2024	Tetsu MORISHIMA	228963	3163

25944 7590 09/21/2026

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EXAMINER	
WONG, ERIC K	

ART UNIT	PAPER NUMBER
2874	

NOTIFICATION DATE	DELIVERY MODE
09/21/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/698,915	Applicant(s) MORISHIMA et al.	
	Examiner Eric K Wong	Art Unit 2874	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 9/1/2026.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-8 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-2 and 4-8 is/are rejected.

8) ☑ Claim(s) 3 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Continued Examination Under 37 CFR 1.114

1. A request for continued examination under 37 CFR 1.114, including the fee set forth in 37 CFR 1.17(e), was filed in this application after final rejection. Since this application is eligible for continued examination under 37 CFR 1.114, and the fee set forth in 37 CFR 1.17(e) has been timely paid, the finality of the previous Office action has been withdrawn pursuant to 37 CFR 1.114. Applicant's submission filed on 9/1/2026 has been entered.

Response to Arguments

2. Applicant's arguments, filed 8/12/2026, have been fully considered and are persuasive. Therefore, the rejection has been withdrawn. However, upon further consideration, a new ground(s) of rejection is made in view of US 2003/0215191 to Inagaki et al (which was cited in the first office action). It appears that the contact surfaces and the recess are opposite of what is claimed. Although Shimikawa discloses a protrusion where the recess should be, rearranging this recess towards the center appears to be a mere rearrangement of shapes or profiles. Inagaki on the other hand discloses a round profile with two contact points with a recess therebetween. Despite lacking two parallel points that define a square-type profile, it appears this feature also falls into a change in shape obviousness rationale. Therefore, an updated rejection follows below.

Claim Rejections - 35 USC § 103

3. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

4. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

5. Claim(s) 1-2 and 4-7 is/are rejected under 35 U.S.C. 103 as being unpatentable over Shimikawa (cited in prior action) in view of US 2003/0215191 to Inagaki et al.

Shimikawa discloses in the abstract and figures, an optical connector comprising:

- a glass fiber (F) and a resin coating covering the glass (film F2),
- a front end portion (towards F1 in figure 8);
- a rear end portion (end towards 7 in figure 8) located on a side opposite to the front end portion (figure 8); and
- a ferrule assembly (2) having a ferrule fixed to a tip end portion (2a) and a holding portion (23; figure 9) to which a rear end portion of the ferrule is fixed and which is provided with a flange (3);
- a housing having an inner wall surface that defines an internal space in which at least the holding portion of the ferrule assembly is accommodated, a flat surface that constitutes a part of the inner wall surface and is provided at a position facing a part of an outer peripheral surface of the flange, and a positioning portion that limits movement of the holding portion in the internal space (figures 11-12 show how the holding portion secures the ferrule in the housing); and
- an elastic member (5) that biases the flange toward the positioning portion,
- wherein the reference surface includes a contact part (surface 23d in figure 11) provided apart from each other by a predetermined distance (the area bounded by

spring 23k) and restricting a rotational direction of a flange as a result of the planar surface of the flange being brought into parallel contact with the planar surface of the inner wall surface with the planar surface of the flange contacting the contact part (when the flange is assembled with the provided spring cutout portion that resides within a channel that creates two contact points, the flange cannot rotate).

As to claim 2, the reference surface has a recessed part (23k) between the two flat reference parts.

As to claim 4, the claim recites a general "floating" terminology defined as the position changes as the ferrule is pushed through. Figure 8 shows some degree of movement and the spring allows for movement in the longitudinal direction.

As to claim 5, the outer shape of the flange is quadrangular (figure 11).

As to claim 6, an outer portion is curved (unlabeled corners are curved in figure 11).

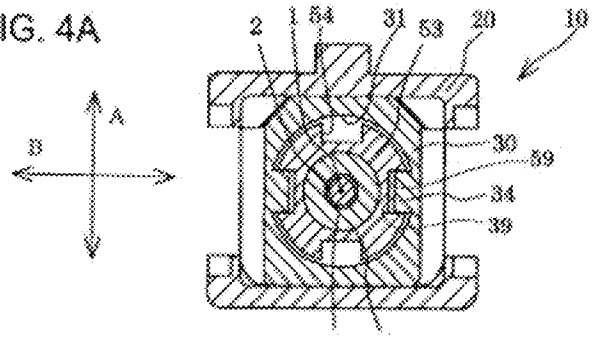
As to claim 7, a multicore fiber is disclosed (paragraph 87).

As to claim 8, the "line" extending along the reference surface coincides with the flange (figure 11).

However, as mentioned in the Response to Arguments above, Shimikawa is a square-type profile but the recess is not located between two contact points. Moving the shapes or recess portions would involve routine skill in the art.

Inagaki discloses a round-type profile having a recess between two contact points (figure 4A).

FIG. 4A



It would have been obvious to one having ordinary skill in the art to swap the recess portion to be between two contact points as taught in Tiara in the square-type shape of Shimikawa as a design choice for a change in size, shape, rearrangement or reversal of parts. *In re Japikse*, 86 USPQ 70, *In re Rose*, 105 USPQ 237 (CCPA 1955), *In re Einstein*, 8 USPQ 167.

Allowable Subject Matter

6. Claim 3 is objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims. The prior art fails to explicitly disclose the projection that protrudes as claimed.

Conclusion

7. Any inquiry concerning this communication or earlier communications from the examiner should be directed to Eric K Wong whose telephone number is (571)272-2363. The examiner can normally be reached M-Tu, Th-F 8A-6P.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Thomas Hollweg can be reached at 571-270-1739. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

ERIC K. WONG
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